

Can SHAPE data discriminate RNA 3D structures? (CASP17 R2307)

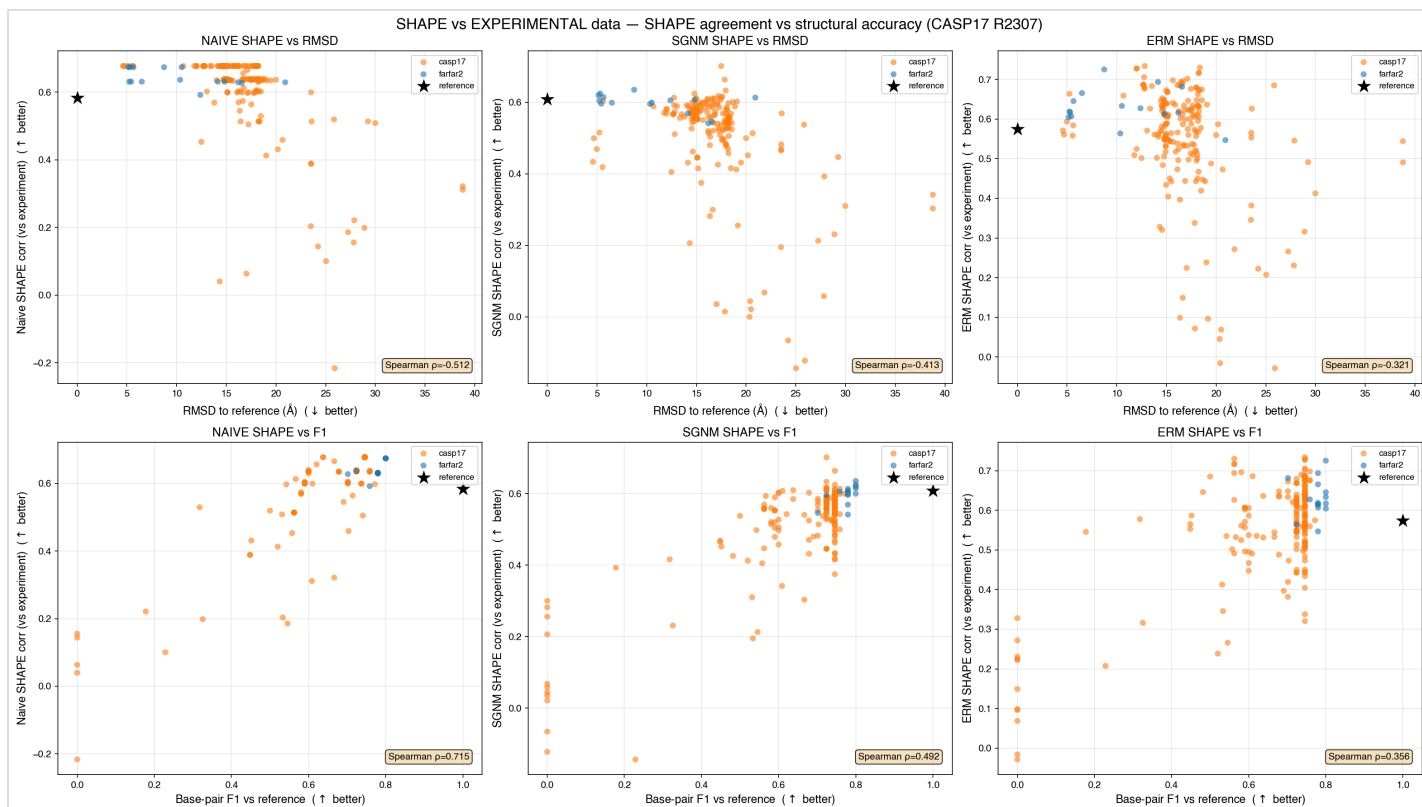
CASP17 target ****R2307**** (reference: FARFAR2 `Mol9\_reference\_UtoG\_buildloop`). Decoys: FARFAR2 + CASP17 submitted models. Can SHAPE reactivity fish low-RMSD models from a mostly-bad pool?

Bottom line — decoy-retrieval metrics

predictor	best_top5	best_top10	auprc	auprc_ci95	auprc_p	EF_top10pct
naive (vs expt)	12.01	10.68	0.072	[0.038, 0.216]	0.093	0.92
SGNM (vs expt)	8.74	5.15	0.141	[0.045, 0.329]	0.054	2.75
ERM (vs expt)	8.74	8.74	0.061	[0.031, 0.099]	0.559	0.0
naive (vs ref)	5.32	5.15	0.193	[0.078, 0.410]	0.005	2.77
SGNM (vs ref)	4.69	4.69	0.097	[0.036, 0.250]	0.1755	1.84
ERM (vs ref)	5.15	5.15	0.192	[0.032, 0.458]	0.0075	2.75

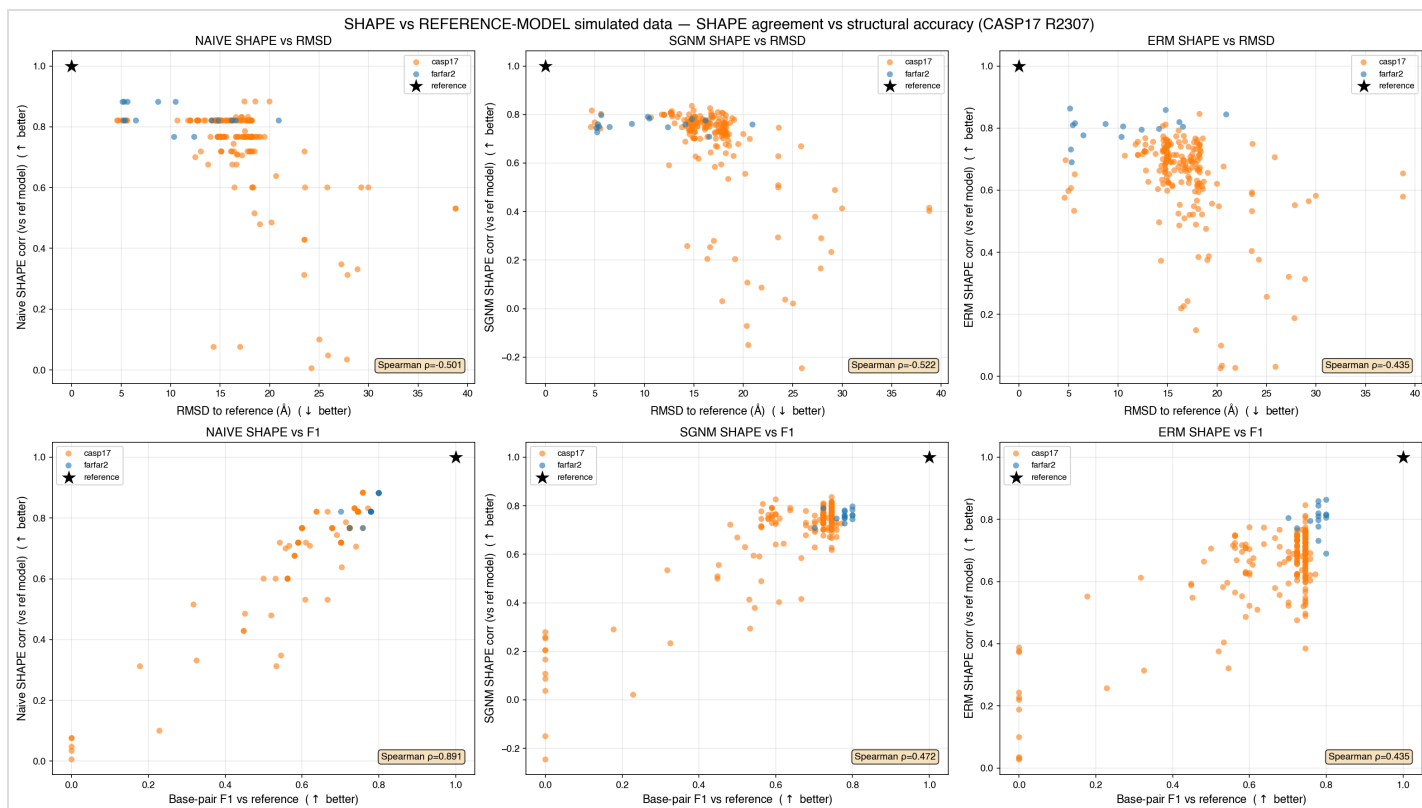
- This is a ****retrieval**** problem: positive = RMSD < 6 Å (11 of ~203 models). Metrics: best RMSD recovered in the top-k SHAPE-ranked models, AUPRC (random ~ 0.054), and enrichment factor at top 10%.
- ****vs experiment, SGNM is the best discriminator**** (AUPRC 0.14, enrichment 2.8x, best-in-top-10 = 5.2 Å). Naive barely beats random; ERM, despite being the stronger sequence-validated model, does not beat SGNM here (AUPRC ~ random).
- ****The ceiling is modest****: even the true structure's predicted SHAPE only correlates ~0.57-0.61 with experiment, so no model can discriminate sharply.
- Only ~11 positives -> wide CIs; this is ****one target****. GPU-run GNM matched the Mac CPU run exactly (max diff 0.0).

SHAPE vs EXPERIMENTAL data



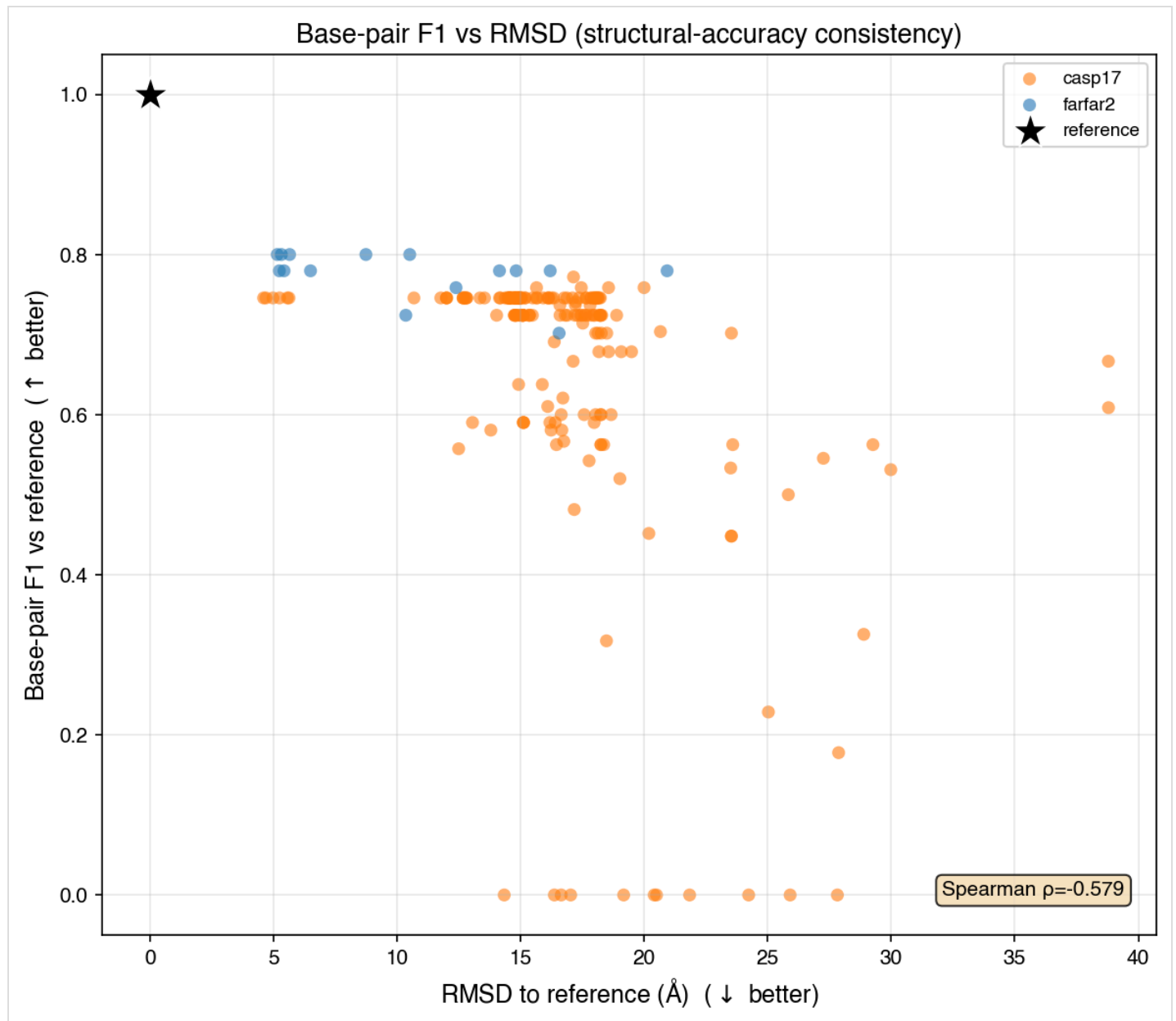
Predicted SHAPE (naive / SGNM / ERM) correlated to experimental 2A3 SHAPE (OpenKnot gRNAde P20). Black star = reference (true) structure at its real correlation (~ 0.6), not 1.0.

SHAPE vs REFERENCE-MODEL simulated data



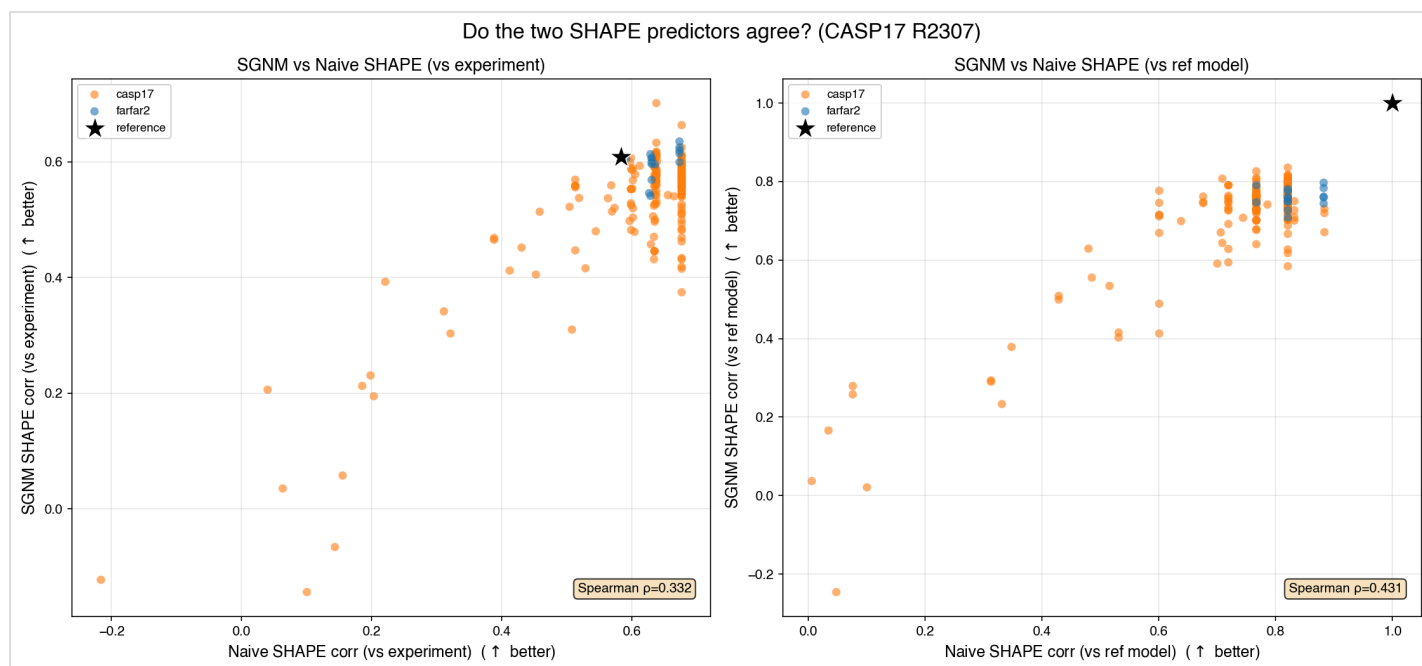
Predicted SHAPE vs the reference-model simulated SHAPE (ceiling analysis); reference trivially sits at 1.0.

Structural-accuracy consistency: base-pair F1 vs RMSD



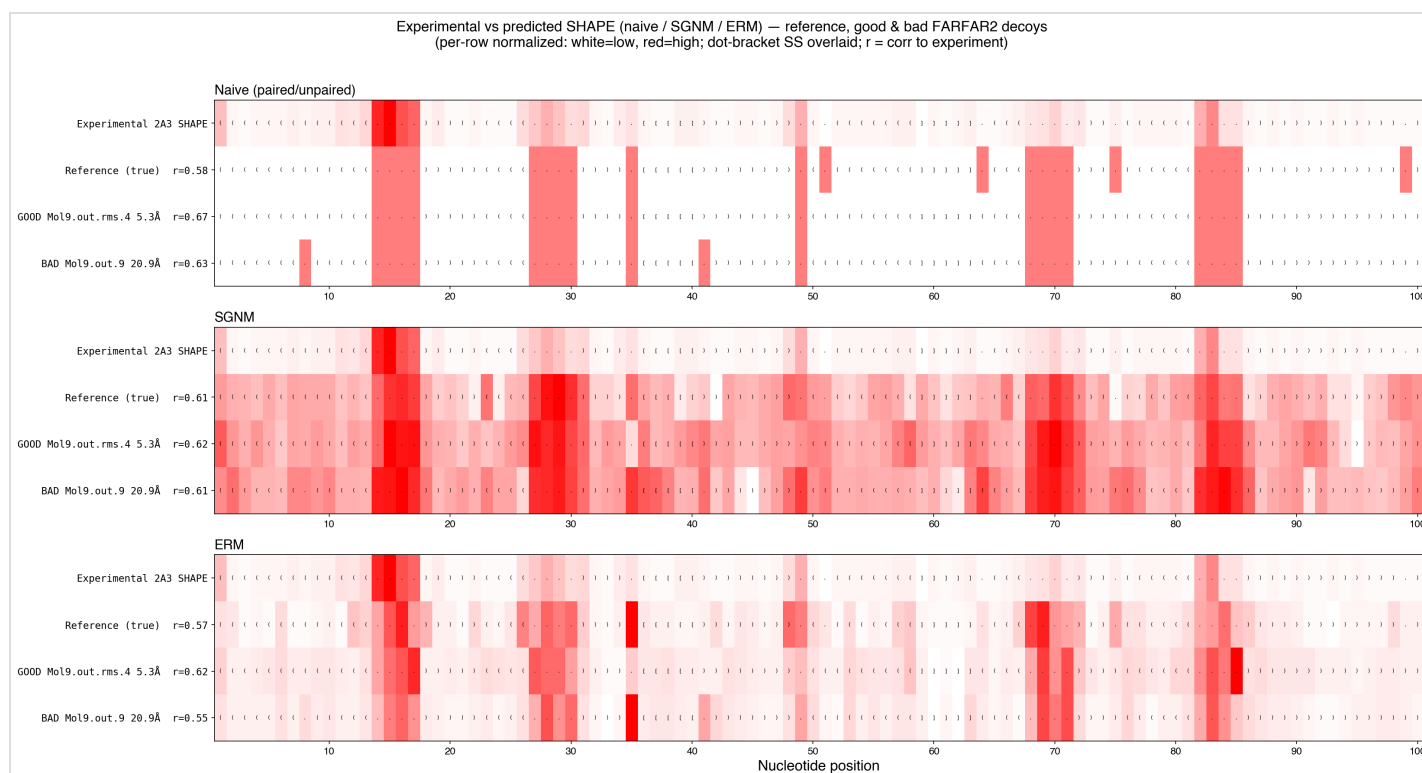
The two structural-accuracy measures agree (lower RMSD $\rightarrow$ higher F1).

Do the two SHAPE predictors agree?



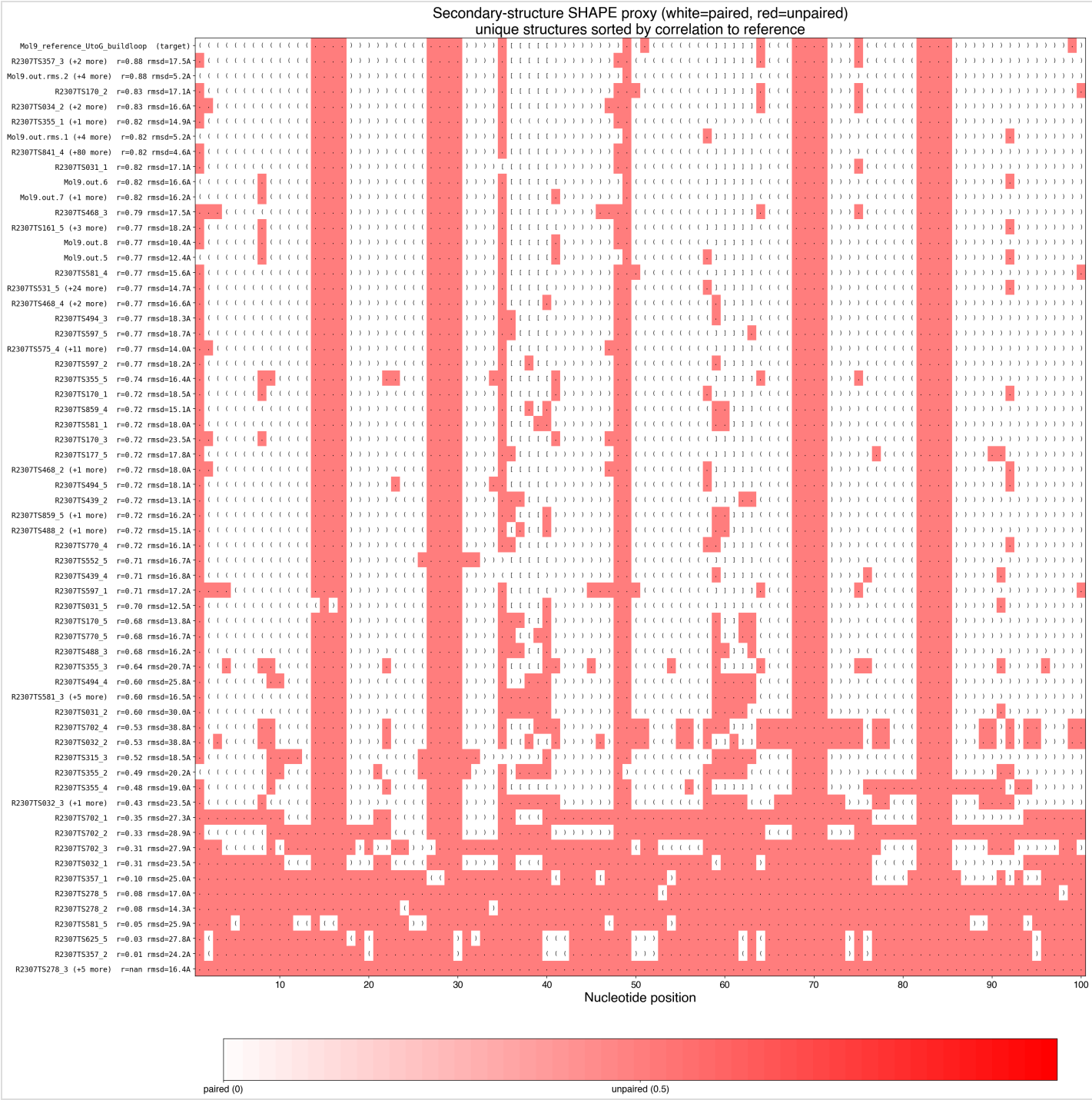
SGNM vs naive SHAPE correlation, for experimental and reference targets.

Focused comparison: experimental vs naive / SGNM / ERM



Experimental 2A3 plus each predictor for the reference, the best GOOD (5.3 Å) and best BAD (20.9 Å) FARFAR2 decoys; dot-bracket SS overlaid.

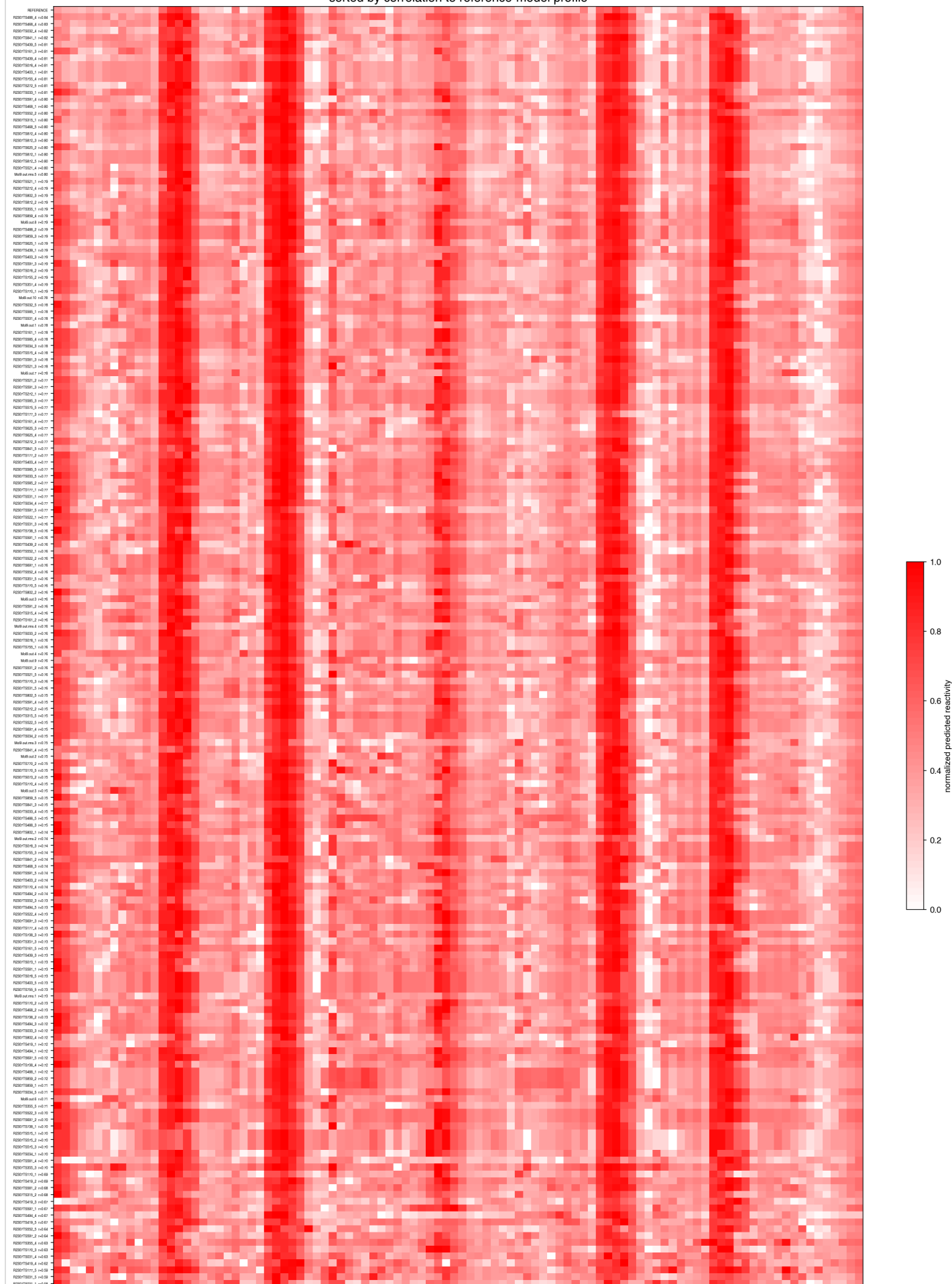
Secondary-structure SHAPE proxy heatmap

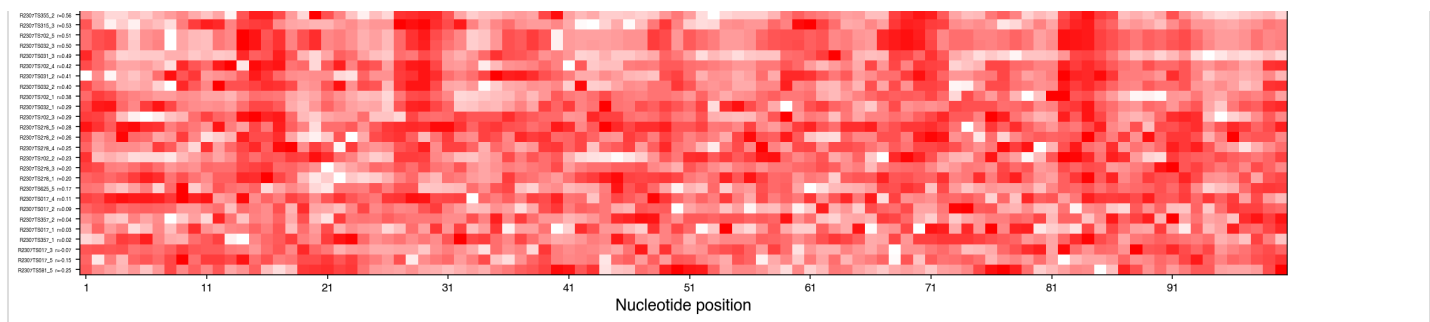


Unique DSSR secondary structures (dot-bracket on each cell), white=paired, light-red=unpaired (0.5), sorted by correlation to the reference.

SGNM-predicted SHAPE profile heatmap

SGNM-predicted SHAPE profiles (per-row min-max normalized)
sorted by correlation to reference-model profile





Continuous SGNM-predicted reactivity (per-row normalized), sorted by correlation to the reference-model profile.

Methods

- ****RMSD****: C1' RMSD to reference via TMscore.
- ****Naive SHAPE****: DSSR paired/unpaired (0 / 0.5), Pearson r vs target.
- ****SGNM / ERM SHAPE****: hmlair/sgnm GNM (CPU) and equivariant (GPU) models predict per-residue reactivity from 3D structure; Pearson r vs target.
- ****Base-pair F1****: F1 of DSSR base pairs vs reference.
- Targets: 'vs experiment' = OpenKnot 2A3 data; 'vs reference' = SHAPE predicted for the reference structure (ceiling).